Лопиталь

$$1)\ \lim_{x\to 0}\frac{2x^2-x}{\ln(1-5x)}=\left\{\frac{0}{0}\right\}=\lim_{x\to 0}\frac{4x-1}{-\frac{5}{1-5x}}=-\frac{1}{-5}=\frac{1}{5}$$

$$2)\lim_{x\to\frac{\pi}{4}}\frac{\cos(2x)}{1-\mathrm{tgx}}=\left\{\frac{0}{0}\right\}=\lim_{x\to\frac{\pi}{4}}\frac{-2\sin(2x)}{-\frac{1}{\cos^2 x}}=\lim_{x\to\frac{\pi}{4}}2sin(2x)\cdot cos^2x=$$

$$=2\cdot sin\frac{\pi}{2}\cdot cos^2\frac{\pi}{4}=2\cdot 1\cdot\frac{1}{2}=1$$

$$3)\ \lim_{x\to 0}(cosx)^{-\frac{1}{x^2}}=\left\{1^{\infty}\right\}=\lim_{x\to 0}e^{ln(cosx)^{-\frac{1}{x^2}}}=\lim_{x\to 0}e^{-\frac{1}{x^2}ln(cosx)}=$$

$$=\lim_{x\to 0}e^{-\frac{ln(cosx)}{x^2}}=e^{\lim\limits_{x\to 0}-\frac{ln(cosx)}{x^2}}=\{e^{-\frac{0}{0}}\}=$$

$$=e^{\lim\limits_{x\to 0}-\frac{\frac{-sinx}{cosx}}{2x}}=e^{\lim\limits_{x\to 0}\frac{tgx}{2x}}=e^{\lim\limits_{x\to 0}\frac{x}{2x}}=e^{\frac{1}{2}}=\sqrt{e}$$

Альтернативно : $e^{\lim\limits_{x\to 0}-\frac{ln(cosx)}{x^2}}=e^{\lim\limits_{x\to 0}\frac{ln(1-0.5x^2)}{-x^2}}=e^{\lim\limits_{x\to 0}\frac{-0.5x^2}{-x^2}}=e^{0.5}=\sqrt{e}$

$$4)\ \lim_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{sin^2x}\right)=\{\infty-\infty\}=\lim_{x\to 0}\left(\frac{sin^2x-x^2}{x^2sin^2x}\right)=$$

$$=\lim_{x\to 0}\left(\frac{\left(\left(x+\frac{x^3}{6}\right)+o(x^3)\right)^2-x^2}{x^2\left(x+\frac{x^3}{6}+o(x^3)\right)^2}\right)=\lim_{x\to 0}\left(\frac{\frac{1}{3}x^4+\frac{x^6}{36}+o(x^6)}{x^2\left(x^2-\frac{1}{3}x^4+\frac{x^6}{36}+o(x^4)\right)}\right)=$$

$$=\lim_{x\to 0}\frac{\frac{1}{3}x^4+o(x^4)}{\left(x^4+o(x^6)\right)}=\text{потому что } o(x^6)\text{ в нуле уж точно является } o(x^4)=$$

$$=\lim_{x\to 0}\frac{\frac{1}{3}x^4+o(x^4)}{\left(x^4+o(x^4)\right)}=\lim_{x\to 0}\frac{\frac{1}{3}+o(1)}{(1+o(1))}=\frac{1}{3}$$

$$sinx=x+\frac{x^3}{6}+o(x^3)$$

$$\text{ЧИСЛИТЕЛЬ}=\left(\left(x+\frac{x^3}{6}\right)+o(x^3)\right)^2-x^2=\left(x+\frac{x^3}{6}\right)^2+2\left(x+\frac{x^3}{6}\right)\cdot o(x^3)+(o(x^3))^2-x^2=$$

$$=\frac{1}{3}x^4+\frac{x^6}{36}+2x\cdot o(x^3)+\frac{1}{3}x^3\cdot o(x^3)+o(x^6)=\frac{1}{3}x^4+\frac{x^6}{36}+o(x^4)$$

5) Для вас задачка :

Найдите коэффициент при x^4 в разложении $f(x)=e^x\cdot\cos(6x)$

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